Shriners Standard Foot Model

Engineering Manual

Table of Contents

Biomechanical Model – Construction and Definition

1.1: Model Construction – Marker Names, Placements, and Considerations

<u>Marker Names and Placements</u>

The Shriners Standard Foot Model defines markers as anatomic (**Table 1.1.1:** ^{A}) and/or technical (**Table 1.1.1:** ^{T}), according to the following usages. Markers are also described in the anatomic and technical coordinate system definitions.

> **Anatomic marker:** resides at a specific, palpable anatomic landmark with known critical alignment parameters – used in a static calibration trial to define the orientation of the marker's parent segment with respect to the global (lab) space.
>
> **Technical marker:** resides at a predefined location on a given segment – used in a dynamic trial to reconstruct anatomic coordinate systems for the given segment. Although specific landmarks are not required, the authors have provided descriptions of optimized locations to limit soft tissue artifact, optimize visibility, assist with symmetry, and reduce collinearity.

NOTE: An anatomic marker that is only present in the static trial is termed a calibration marker. In the naming convention employed, calibration markers are given a '.Cal' extension to signify their absence in the dynamic workflow.

NOTE: where possible, certain markers are defined as used in both anatomic and technical coordinate system registration – denoted in Table 1.1.1 as ^{A,T}.

Table 1.1.1: The Shriners Standard Foot Model consists of the following basic marker set. Superscripts indicate markers that are anatomical[A] and/or technical[T].

Marker Name	Landmark	Location	Critical Alignment
Left_Lateral_Malleolus, Right_Lateral_Malleolus [A,T]	Fibula lateral malleolus	Lateral malleoli so that marker center lies on ankle flexion/extension axis.	Anterior/Posterior; Superior/Inferior
Left_Medial_Malleolus.Cal, Right_Medial_Malleolus.Cal [A]	Tibia medial malleolus	Medial malleoli so marker center lies on ankle flexion/extension axis.	Anterior/Posterior; Superior/Inferior
Left_Posterior_Calcaneus, Right_Posterior_Calcaneus[T]	Posterior calcaneus	Medial/lateral center of posterior calcaneus	Medial/Lateral; directly in the middle of the heel
Left_Lateral_Calcaneus, Right_Lateral_Calcaneus[T]	Lateral calcaneus	Lateral calcaneus, superior to bulge in heel pad	Anterior/Posterior alignment should be level with the Medial_Calcaneus
Left_Medial_Calcaneus, Right_Medial_Calcaneus[T]	Medial calcaneus	Medial calcaneus, superior to bulge in heel pad	Anterior/Posterior alignment should be level with the Lateral_Calcaneus
Left_Peroneal_Trochlea.Cal, Right_Peroneal_Trochlea.Cal[A]	Peroneal Trochlea	Directly on peroneal trochlea of the lateral calcaneus	Superior/Inferior, Anterior/Posterior *Only critical if used to calculate calcaneal pitch*
Left_1st_MT_Base, Right_1st_MT_Base[T]	Base of 1st metatarsal	Dorsal aspect of 1st metatarsal base, avoiding flexor hallucis longus tendon	None, technical only
Left_1st_MT_Head, Right_1st_MT_Head[T]	Head of 1st metatarsal	Dorsal aspect of 1st metatarsal head, avoiding flexor hallucis longus tendon	None, technical only
Left_1st_MT_Medial_Base.Cal, Right_1st_MT_Medial_Base.Cal[A]	Base of 1st metatarsal	Medial aspect of base of 1st metatarsal	Superior/Inferior
Left_1st_MT_Medial_Head.Cal Right_1st_MT_Medial_Head.Cal[A]	Head of 1st metatarsal	Medial aspect of head of 1st metatarsal	Superior/Inferior
Left_2nd3rd_MT_Base.Cal Right_2nd3rd_MT_Base.Cal[A]	Gap between the 2nd and 3rd metatarsal bases	Dorsal aspect of forefoot between 2nd and 3rd metatarsal bases	Medial/Lateral (between 2nd and 3rd rays)

Left_2nd3rd_MT_Head.Cal Right_2nd3rd_MT_Head.Cal[A]	Gap between the 2nd and 3rd metatarsal heads	Dorsal aspect of forefoot between 2nd and 3rd metatarsal heads (above metatarsal-phalangeal joints)	Medial/Lateral (between 2nd and 3rd rays)
Left_5th_MT_Head, Right_5th_MT_Head[T]	Head of 5th metatarsal	Dorsal aspect of 5th metatarsal head, avoiding flexor digitorum longus tendon	None, technical only
Left_1st_MTP_Joint.Cal Right_1st_MTP_Joint.Cal[A]	1st metatarsal-phalongeal joint	Dorsal aspect of 1st metatarsal head, just proximal to the MTP joint line	Medial/Lateral
Left_Hallux, Right_Hallux[T]	Nail of hallux	Middle of the nail, aligning with long axis of the hallux	Medial/Lateral

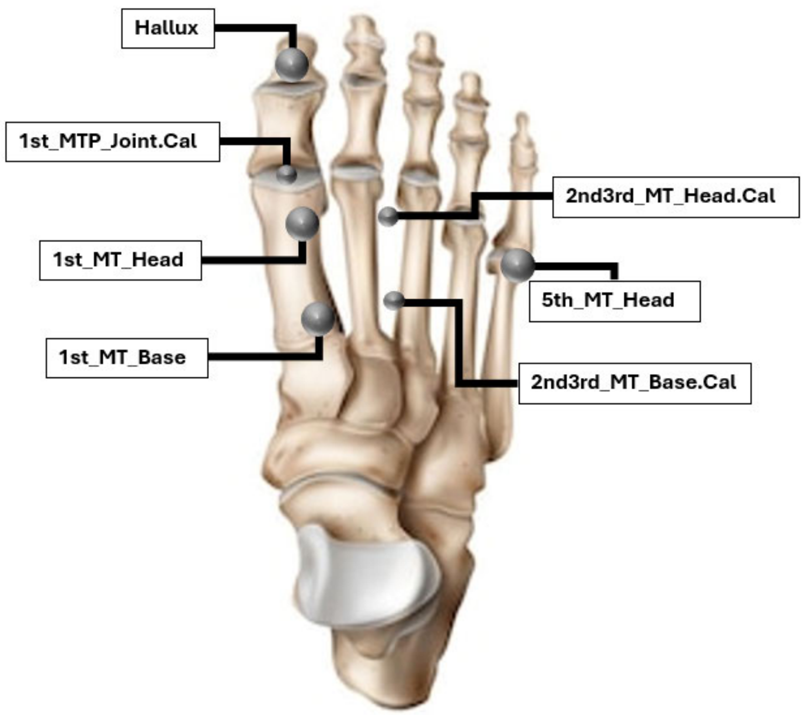


Figure 1.1.1: Top-down view of foot model marker placements

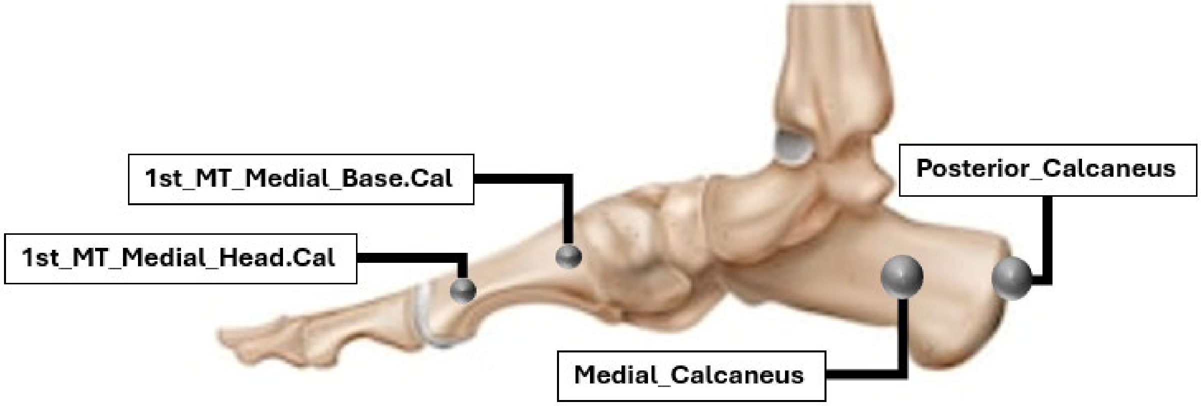


Figure 1.1.2: Medial view of foot model marker placements

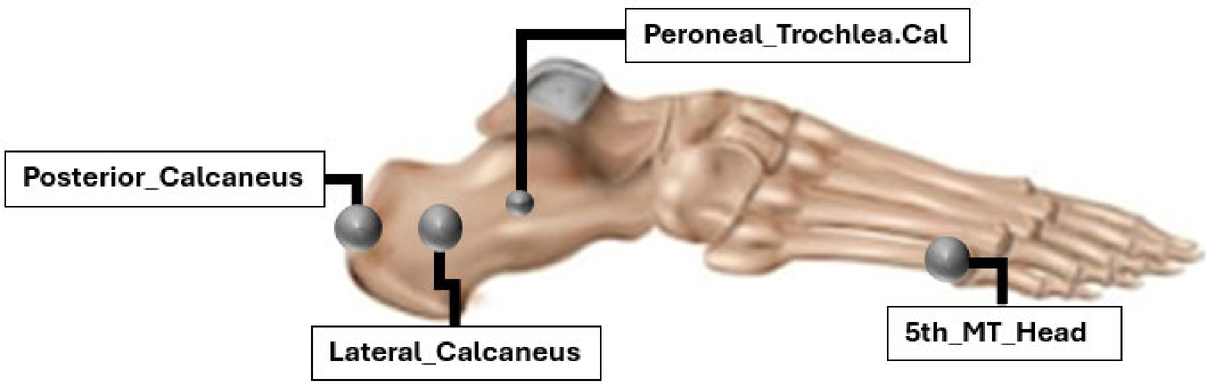


Figure 1.1.3: Lateral view of foot model marker placements

<u>Marker Placement Notes</u>

1st_MT_Head | *1st_MT_Base*

- Try to avoid the flexor hallucis longus tendon
- Do not want them moving relative to each other during the dynamic trial
- May need to move them to the lateral side of the 1st Metatarsal to avoid tendon

2nd3rd_MT_Head.Cal | *2nd3rd_MT_Base.Cal*

- Calibration markers only, does not matter if they are on a tendon
- The direction of these markers (Base to Head) should align with clinical impression of forefoot adduction/abduction

1st_MTP_Joint.Cal | Hallux

- Markers should be placed such that a line connecting the *1st_MTP_Joint.Cal* and *Hallux* markers is parallel with the long axis of the hallux

Calcaneus Markers

- Posterior marker should be directly in the center of the heel (medial/lateral direction)
- Medial and lateral markers should be symmetric (medial/lateral direction) relative to the posterior marker
- Medial and lateral markers should be placed as far anteriorly as possible to maximize the length of the vector between posterior marker and their midpoint
- (See simple image below for placement emphasis)

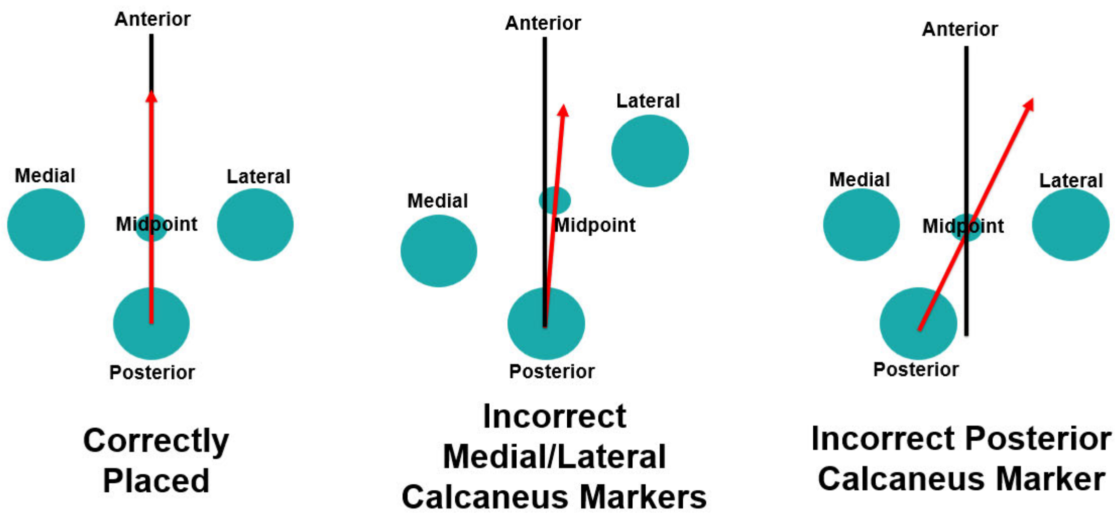


Figure 1.1.4: The black line represents the ideal vector for the hindfoot. The red arrow represents the actual vector based on alignment of markers. Note that incorrect placement of the medial and lateral calcaneus markers (*middle)* can cause some shift, but the greatest shift is caused by incorrect posterior calcaneus placement (*right*).

1st_MT_Medial_Head | 1st_MT_Medial_Base

- Calibration markers only
- The direction of these markers (Base to Head) should align with clinical impression of forefoot pitch
- If aligned with forefoot inclination, markers can be moved up and down along parallel lines to view them better during the static trial
- Try to maximize distance between these markers to increase vector size and reduce small vector errors

1.2: Model Definition – Segmental Coordinate Systems

The coordinate systems of each segment of interest in the Shriners Standard Foot Model are described here, including all virtual point calculations and relevant axis definitions. Coordinate system axes are defined using general anatomic orientations (medial/lateral, anterior/posterior, superior/inferior). X, Y, Z coordinate labels are provided as additional shortcut nomenclature to aid in defining certain calculation orders. X, Y, and Z axes have the directionality of anterior, leftward, and superior, respectively.

For the figures provided, the following standard color key is employed for Axes:

- Axis 1: 
- Axis 2: 
- Axis 3: 

For the figures provided, the following standard color key is employed for defining lines (where applicable):

- Line 1: 
- Line 2: 
- Line 3: 

Physical markers are shown in silver  and virtual markers are shown in gold 

The Shriners Standard Foot Model segment definitions require the foot to be FULLY plantigrade during static calibration. The hindfoot must be plantigrade to correctly represent the sagittal plane pitch, while coronal plane varus/valgus is accounted for by measuring the angle of the hindfoot against the floor. The forefoot must be plantigrade to correctly represent the coronal and sagittal planes and properly estimate pronation and supination. If the patient cannot achieve a plantigrade position during a standing calibration, instructions for an alternative two-step static using a seated position are provided in Appendix A.

<u>Hindfoot – Anatomic</u>

<u>Description</u>

The hindfoot's anatomical coordinate system is described by three physical markers on the calcaneus (*Posterior_Calcaneus, Lateral_Calcaneus, Medial_Calcaneus)*. *CCAL* is defined as the midpoint between the *Lateral_Calcaneus* and *Medial_Calcaneus* markers. The projection of a vector from *Posterior_Calcaneus* to *CCAL* onto the plane associated with the floor defines the anterior/posterior axis of the hindfoot. The medial/lateral axis is the cross product of the global vertical reference and the anterior/posterior axis. The superior/inferior axis is the cross product of anterior/posterior axis and the medial/lateral axis.

Virtual Points

- *CCAL: CCAL = (Lateral_Calcaneus + Medial_Calcaneus)/2*

Vector	Description	Calculation	Direction	Label
Line 1	Posterior calcaneus to midpoint between lateral and medial calcaneus	*CCAL - Posterior_Calcaneus (Result projected to floor)*		
Axis 1	Line 1 with calcaneal pitch applied to it		+Anterior	X-Axis
Line 2	Global vertical axis rotated by global hindfoot varus/valgus angle about Axis 1			
Axis 2		Line 2 x Axis 1 (X)	+Medial	Y-Axis
Axis 3		Axis 1 (X) x Axis 2 (Y)	+Superior	Z-Axis

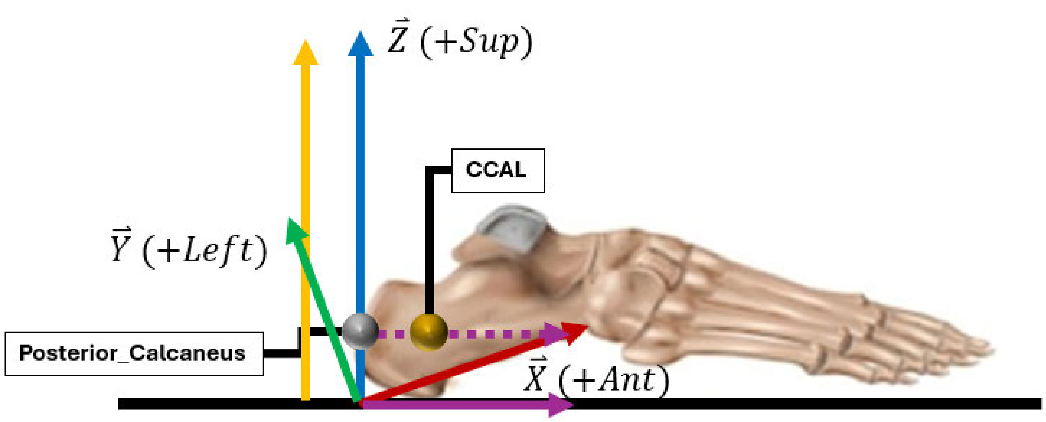


Figure 1.2.1: The hindfoot anatomical coordinate system from a lateral view. Line 1 (purple) is projected to the floor and then adjusted based on the calcaneal pitch to create Axis 1 (red).

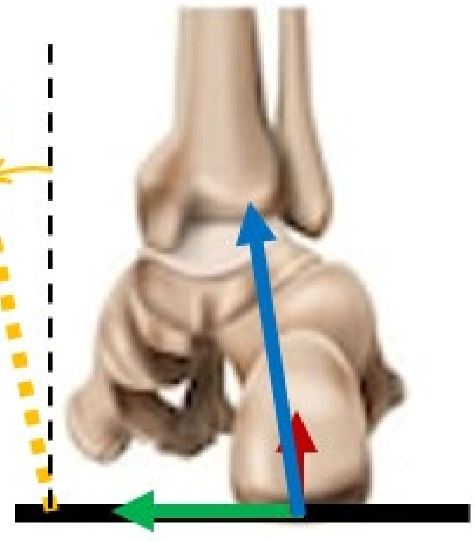


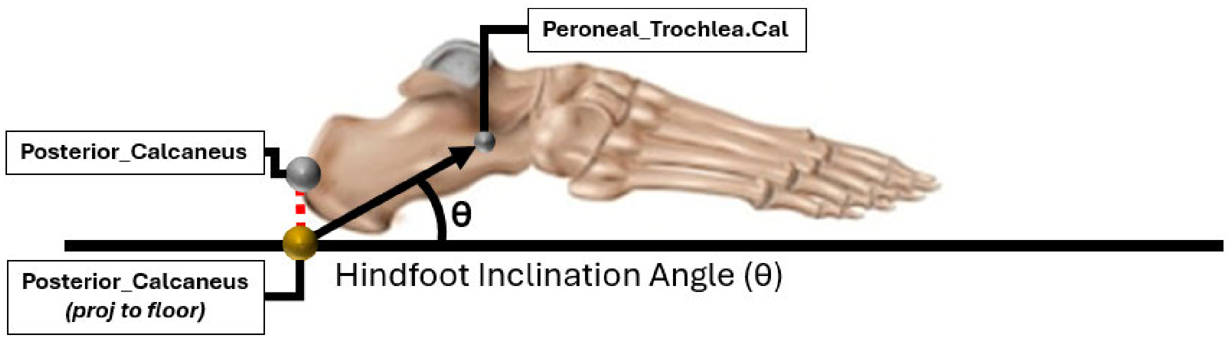
Figure 1.2.2: The hindfoot anatomical coordinate system from a posterior view. Line 2 (yellow) represents the global vertical axis rotated by the global hindfoot varus/valgus angle.

Figure 1.2.3: The hindfoot anatomical coordinate system from a superior view. Axis 1 (red) defines hindfoot progression.

Multiple options exist for estimating calcaneal pitch (Relevant to Axis 1):

Option 1: Calcaneal Pitch from Markers

If no radiographic images are available, the *Peroneal_Trochlea.Cal* marker can be used to estimate calcaneal pitch. A vector between the origin of the hindfoot and the projection of *Peroneal_Trochlea.Cal* onto the sagittal plane is created. The angle between this vector and the hindfoot anterior/posterior axis is the hindfoot inclination angle. The hindfoot inclination angle is then used in a regression equation to estimate calcaneal pitch. A non-plantigrade foot will drastically impact this calculation.



$$Calcaneal\ Pitch\ =\ (0.776\ \cdot\ Hindfoot\ Inclination\ Angle) - 11.6$$

Figure 1.2.4: Calcaneal pitch calculation using the marker-based method.

NOTE: This method of calcaneal pitch estimation relies heavily on correct placement of the Peroneal_Trochlea.Cal marker

Option 2: Calcaneal Pitch from Radiograph

Calcaneal pitch can be calculated from lateral weight-bearing foot radiographs, if available. In maintaining the plantigrade assumptions of the model, the angle between the calcaneal inclination axis and the supporting horizontal surface is calculated (1.2.5). In the case of a non-plantigrade radiograph, measurement of the plantar surface should align with the soft tissue of the plantar surface of the foot, not the weight-bearing surface (1.2.6)

Figure 1.2.5: Calcaneal pitch represented on lateral foot radiograph for plantigrade foot

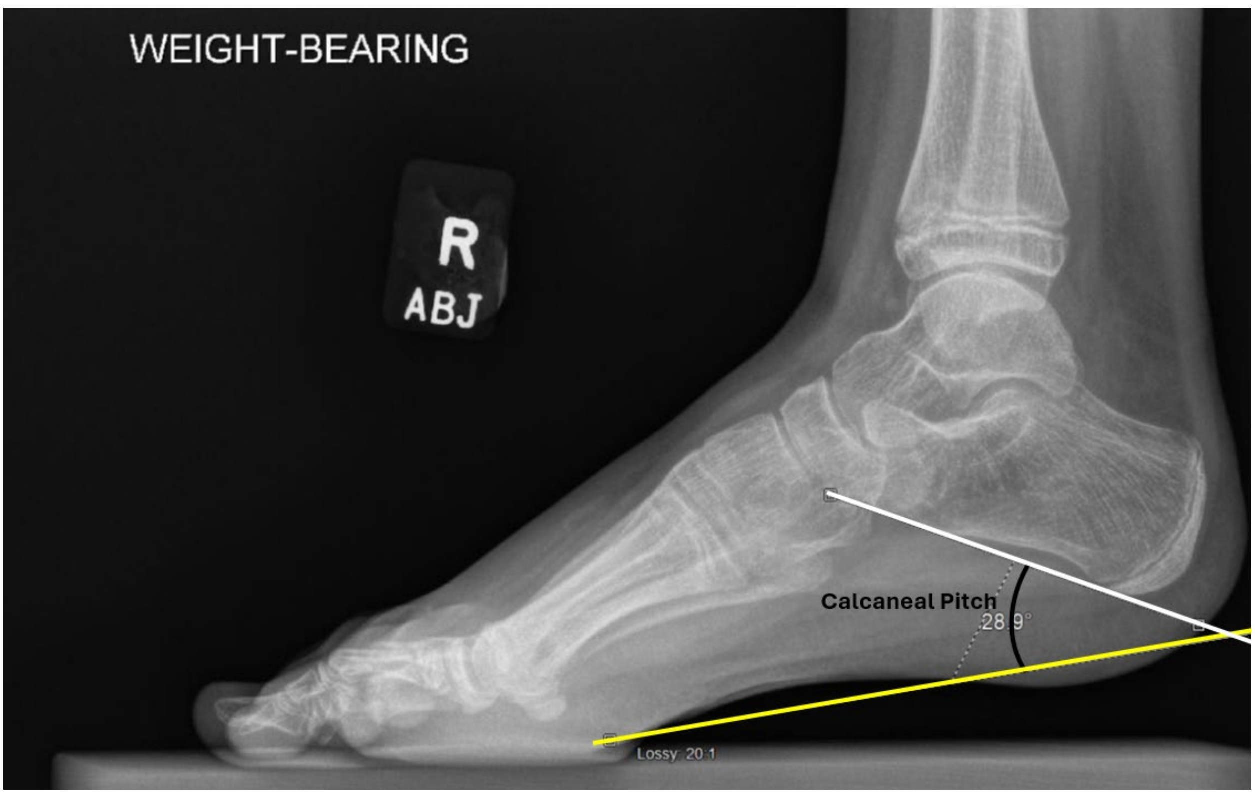


Figure 1.2.6: Calcaneal pitch represented on lateral foot radiograph for non-plantigrade foot. This angle is being used for a plantigrade static trial. This angle is taken against the soft tissue surface of the foot.

Figure 1.2.7: An example of a negative calcaneal pitch represented on lateral foot radiograph for plantigrade foot.

<u>NOTE:</u> This is the recommended method of determining calcaneal pitch for the Shriners Standard Foot Model.

<u>NOTE:</u> Using this method is the one way to avoid the plantigrade requirement. The patient can stand without being plantigrade in the static, but they need to stand in the same way during the radiograph. **In this case, the calcaneal pitch model input from the radiograph needs to be calculated relative to the actual floor, instead of the plantar surface of the foot, as shown below (1.2.8).**

Figure 1.2.8: Calcaneal pitch represented on lateral foot radiograph for non-plantigrade foot. This angle is being used for a non-plantigrade static trial.

Additional Modeling Option for Hindfoot Progression Relative to Bimalleolar Axis (Relevant to Line 1):

This method uses radiographic measurements to define the hindfoot progression relative to the bimalleolar axis (anterior/posterior axis). The bimalleolar axis is defined using the position of radiopaque markers (BBs) on the medial and lateral malleoli in the same position as the gait markers. The hindfoot progression line is drawn to be parallel with the lateral border of the calcaneus. The angle between the two lines is taken, with a relative internal angle considered to be positive and relative external angle considered negative. The example below (1.2.9) shows an internal angle (+).

Figure 1.2.9: Hindfoot progression relative to bimalleolar axis represented on AP foot radiograph.

Hindfoot – Technical

Description

The hindfoot's technical coordinate system is described by three physical markers on the calcaneus (*Posterior_Calcaneus, Lateral_Calcaneus, Medial_Calcaneus). CCAL* is defined as the midpoint between the *Lateral_Calcaneus* and *Medial_Calcaneus* markers.

Virtual Points

- *CCAL: CCAL = (Lateral_Calcaneus + Medial_Calcaneus)/2*

Vector	Description	Calculation
Line 1	Posterior calcaneus to midpoint between lateral and medial calcaneus	*CCAL - Posterior_Calcaneus*
Line 2	Mediolateral vector pointing from right to left using medial and lateral calcaneus markers	± (Lateral_*Calcaneus – Medial_Calcaneus)* *+ for Left, - for Right*

The directionality of Line 2 that defines the hindfoot medialolateral axis is side-dependent and must be adjusted to reflect a consistent leftward-positive orientation (as indicated by the "±").

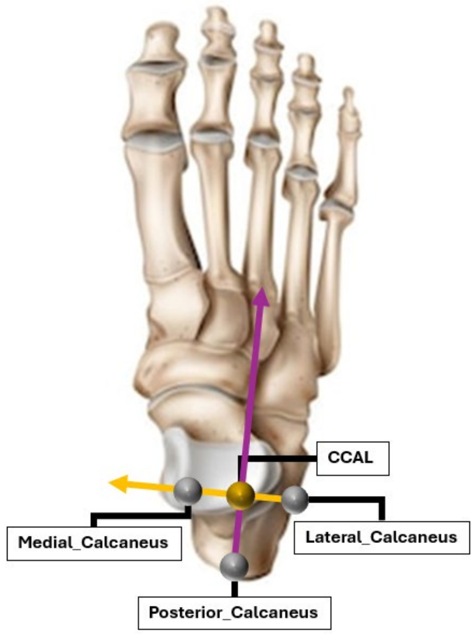


Figure 1.2.10: The hindfoot technical coordinate system is established with three physical markers (medial, lateral, and posterior calcaneus) and one virtual marker (CCAL).

Forefoot - Anatomic

Description

The forefoot's anatomical coordinate system is described by two physical markers (*2nd3rd_MT_Base.Cal, 2nd3rd_MT_Head.Cal*). The projection of a vector from *2nd3rd_MT_Base.Cal* to *2nd3rd_MT_Head.Cal* onto the plantar surface defines the anterior/posterior axis. The medial/lateral axis is the cross product of the global vertical reference and the anterior/posterior axis. The superior/inferior axis is the cross product of anterior/posterior axis and the medial/lateral axis.

Vector	Description	Calculation	Direction	Label
Line 1		*2nd3rd_MT_Head.Cal - 2nd3rd_MT_Base.Cal*		
Line 2		*1st_MT_Medial_Head.Cal - 1st_MT_Medial_Base.Cal*		
Line 3	Global vertical reference			
Axis 1	Line 2 projected onto plane formed by line 1 and line 3		+Anterior	X-Axis
Axis 2		Line 3 x Axis 1 (X)	+Medial	Y-Axis
Axis 3		Axis 1 (X) x Axis 2 (Y)	+Superior	Z-Axis

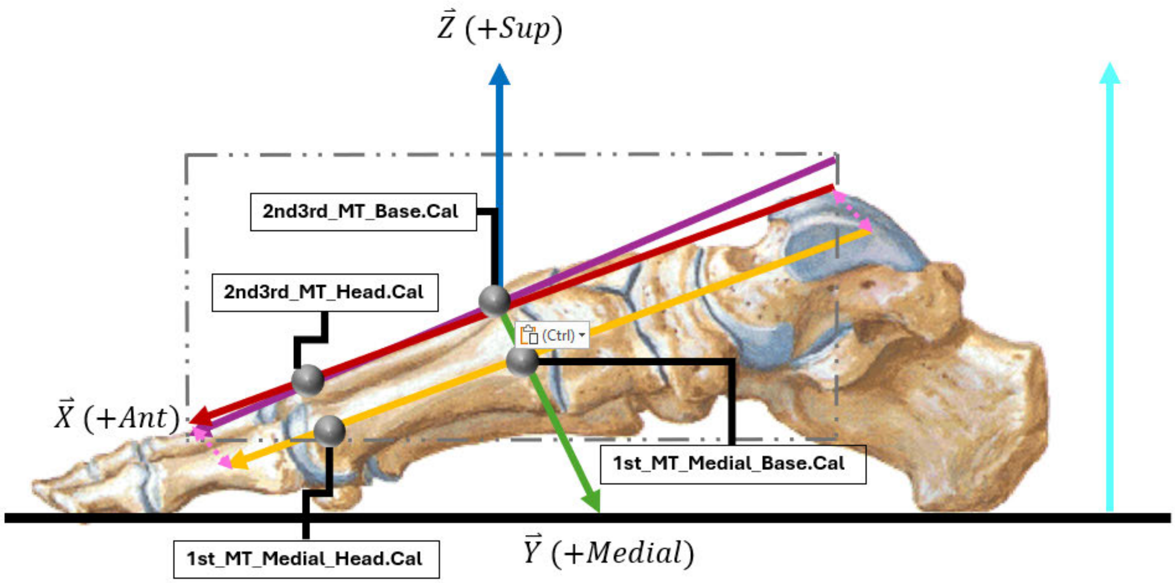


Figure 1.2.11: The forefoot anatomical coordinate system from a medial point of view. Axis 1 (red) is defined by projecting Line 2 (yellow) onto the plane created by Line 1 (purple) and Line 3 (light blue). Line 1 aligns Axis 1 to track forefoot ab/adduction, while Line 2 aligns Axis 1 to track forefoot pitch.

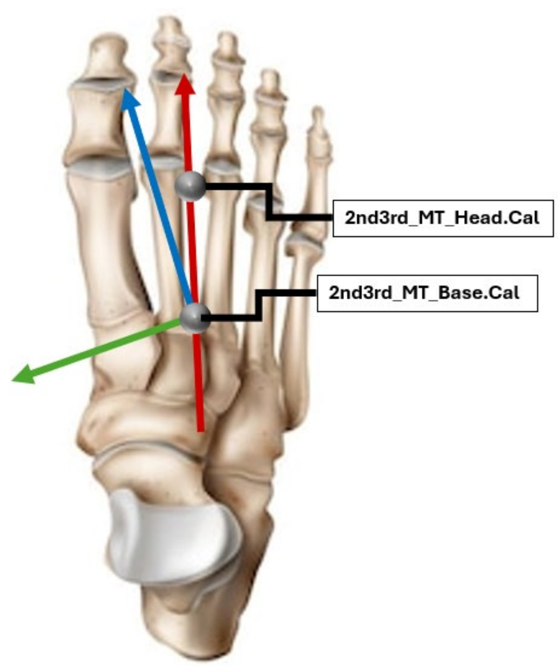


Figure 1.2.12: The forefoot anatomical coordinate system from a superior view. Axis 1 (red) ends up aligned to represent forefoot ab/adduction.

Forefoot inclination can be defined with markers, as described above (Line 2), or measured from radiographs:

Forefoot Pitch from Radiograph (Relevant to Line 1):

Forefoot pitch can be calculated from lateral weight-bearing foot radiographs, if available. The angle between the 1st metatarsal and the ground in the sagittal plane is used for forefoot pitch.

Figure 1.2.13: Forefoot pitch angle represented on lateral foot radiograph for a plantigrade foot.

NOTE: If using forefoot pitch measured from a radiograph, the 1st_MT_Medial_Head.Cal and 1st_MT_Medial_Base.Cal are not required.

NOTE: Overlapping rays on lateral weight-bearing foot radiographs can make it difficult to distinguish the 1st metatarsal. Therefore, use of the *2nd3rd_MT_Base.Cal* and *2nd3rd_MT_Head.Cal* markers is the recommended method of estimating forefoot pitch for the Shriners Standard Foot Model.

Additional Modeling Option for Forefoot Progression Relative to Bimalleolar Axis (Relevant to Line 1):

This method uses Xray measurements to define the forefoot progression relative to bimalleolar axis (anterior/posterior axis). The bimalleolar axis is defined using the position of radiopaque markers (BBs) on the medial and lateral malleoli in the same position as the gait markers. The forefoot progression line is drawn along the long axis of the second metatarsal. The angle between the two lines is taken, with a relative internal angle considered to be positive and relative external angle considered negative. The example below (1.2.14) shows an external angle (-).

Figure 1.2.15: Forefoot progression relative to bimalleolar axis represented on AP foot radiograph.

Forefoot - Technical

Description

The forefoot's technical coordinate system is described by three physical markers (*1st_MT_Base, 1st_MT_Head, and 5th_MT_Head*).

Vector	Description	Calculation
Line 1	1st_MT_Base to 1st_MT_Head	*1st_MT_Head – 1st_MT_Base*
Line 2	Mediolateral vector pointing from right to left using 1st_MT_Head and 5th_MT_Head	Left Foot: *5th_MT_Head – 1st_MT_Head* Right Foot: *1st_MT_Head – 5th_MT_Head*

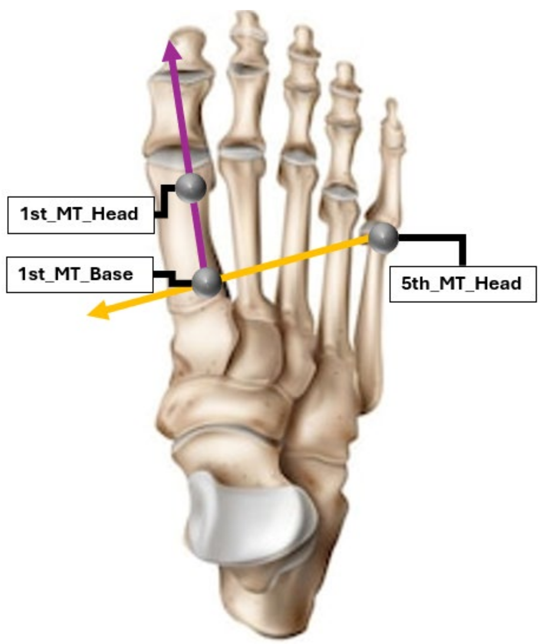


Figure 1.2.16: The forefoot technical coordinate system is established by three physical markers on the forefoot (1st_MT_Base, 1st_MT_Head, and 5th_MT_Head).

Hallux – Anatomic/Technical

The hallux coordinate system is described by three physical markers (*Hallux, 1st_MTP_Joint.Cal, 2nd3rd_MT_Head.Cal*). The anterior/posterior axis is the vector from *1st_MTP_Joint.Cal* to *Hallux*. The superior/inferior axis is the cross product of the anterior/posterior axis and a vector between *1st_MTP_Joint.Cal* and *2nd3rd_MT_Head.Cal* (direction dependent). The medial/lateral axis is the cross product of the superior/inferior axis and anterior/posterior axis.

Vector	Description	Calculation	Direction	Label
Axis 1		*(Hallux - 1st_MTP_Joint.Cal)*	+Anterior	X-Axis
Line 1		*±(2nd3rd_MT_Head.Cal - 1st_MTP_Joint.Cal)* *+ for Left, - for Right*	+Leftward	
Axis 2		Axis 1 (X) x Line 1	+Superior	Z-Axis
Axis 3		Axis 2 (Z) x Axis 1 (X)	+Leftward	Y-Axis

The directionality of Line 2 that defines the hallux flexion axis is side-dependent and must be adjusted to reflect a consistent leftward-positive orientation (as indicated by the "±").

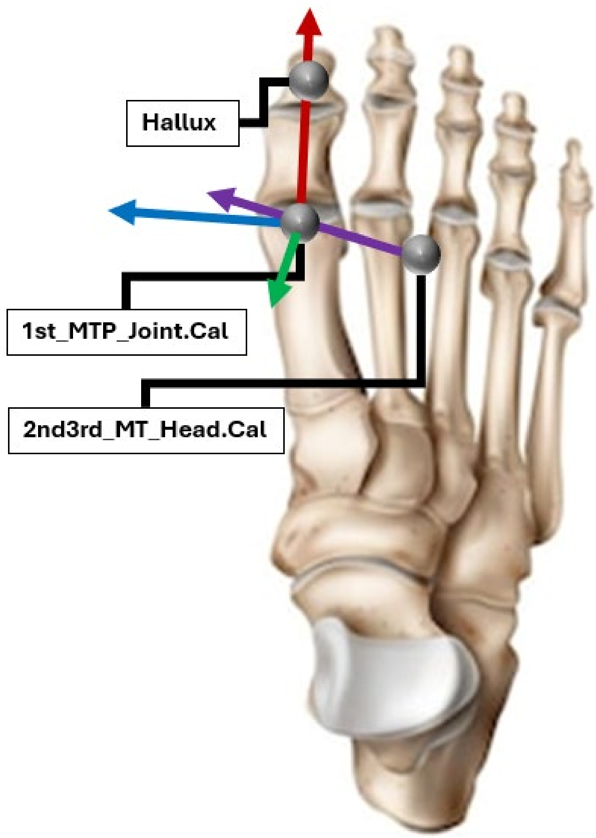


Figure 1.2.17: The hallux anatomical and technical coordinate system from a superior view. The hallux outputs of the model include hallux flexion/extension and hallux varus/valgus.

NOTE: The 1st_MTP_Joint.Cal marker is tracked with the forefoot technical coordinate system

1.3: Model Definition – Cardan Sequences and Joint Angles

The order of the Cardan rotation sequence is expressed in this document by listing the axes from left to right. As an example, a Cardan X-Y-Z rotation sequence is written and may be read thus: rotating the distal segment first about the proximal segment’s X axis, second about the proximal Y axis, and third about the proximal Z axis.

Table 1.3.1: Description of the Shriners Standard Foot Model’s given joints, relevant segments, Cardan order of rotations, sign adjustments, and anatomic angles.

Joint	Distal Segment	Proximal Segment	Cardan Order	Sign Adjustments		Anatomic Angles
				L	R	
Hindfoot wrt Tibia	Hindfoot	Tibia	<Y, X, Z>	-α β -γ	α -β γ	α: Hindfoot (wrt tibia) In(+)/Eversion(-) β: Hindfoot (wrt tibia) Dorsi(+)/Plantar(-)flexion γ: Hindfoot (wrt tibia) Internal(+)/External(-) Rotation
Forefoot wrt Hindfoot	Forefoot	Hindfoot	<Z, X, Y>	-α β -γ	α -β γ	α: Forefoot (wrt tibia) In(+)/Eversion(-) β: Forefoot (wrt tibia) Dorsi(+)/Plantar(-)flexion γ: Forefoot (wrt Hindfoot) Internal(+)/External(-) Rotation
MTP1	Hallux	Forefoot	<Y, X, Z>	-α β -γ	α -β γ	α: Hallux Varus(+)/Valgus(-) β: Hallux Extension(+)/Flexion(-)

1.4: Model Definition – Kinetic Calculations

The Shriners Standard Foot Model does not attempt to provide kinetic calculations for the individual segments. Ankle kinetics are only described using the same methodology as the single segment foot model used in the Shriners Children’s Standard Gait Model (SCGM).

References

Davids JR, Gibson TW, Pugh LI. Quantitative segmental analysis of weight-bearing radiographs of the foot and ankle for children: normal alignment. J Pediatr Orthop. 2005;25(6):769-776. doi:10.1097/01.bpo.0000173244.74065.e4

Kruger, Karen & Fischer, Patrick & Augsburger, Sam & Feng, Jing & Girouard, Jean-François & Gregory, Daniel & Johnson, Logan & MacWilliams, Bruce & Mcmulkin, Mark & Nelson, Bradley & Warshauer, Spencer & Saraswat, Prabhav & Chafetz, Ross. (2024). The Shriners Children's Standard Gait Model (SCGM). Gait & Posture. 114. 10.1016/j.gaitpost.2024.09.007.

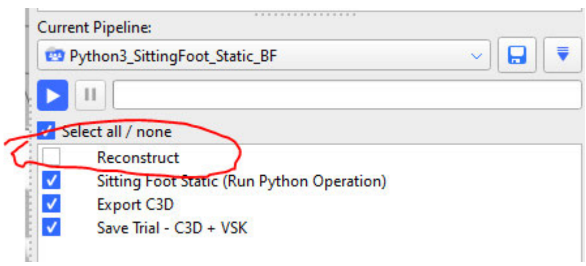
Appendix A: Sitting Static Instructions

Foot Model: When Patient cannot stand foot flat - SPECIAL PROCESSING

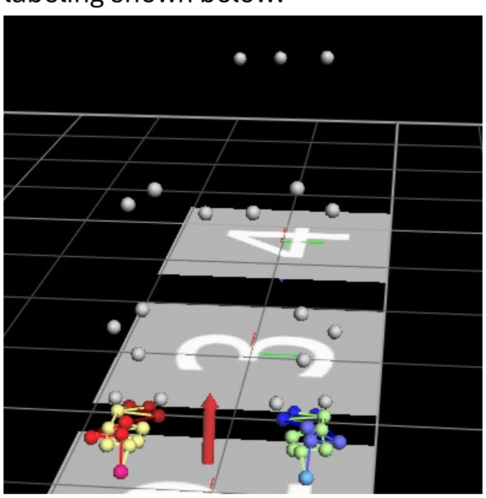
The model relies on plantigrade positioning of the feet (in both dorsi/plantarflexion and inversion/eversion). If a subject is not able to stand foot flat, a special sequence of static trials and processing is needed. A seated static with all foot calibration markers and dynamic foot markers is needed with the feet placed plantigrade to properly process the foot model (no markers above the tibia required). A typical standing static should be obtained for lower extremity modeling, this can be done with or without the foot calibration markers and dynamic foot marker set. Processing should be done live to get allow for quality checking the marker placement but can be done post collection if time is a factor. If doing it post collection, at least confirm that all foot model markers are visible upon reconstruction.

To process, do the following:

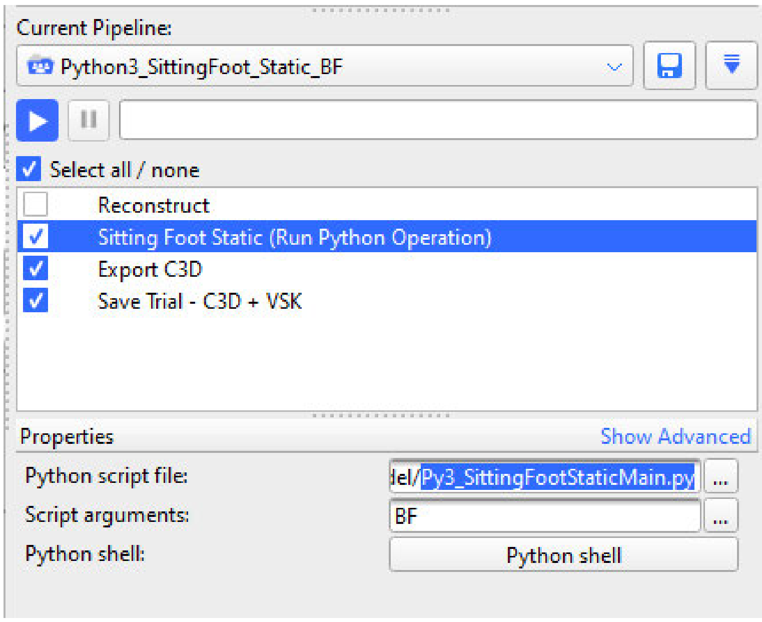
1) Create subject attach “SMACNet FootModel noKAD.vst”.
2) Enter anthropometric data, **including hind foot angles (varus/valgus: taken while subject is seated)**
3) Place all calibration and dynamic foot markers.
    a. Capture trial. Trial description: Sitting Foot Static
    b. Reconstruct (top pipeline command) the sitting static



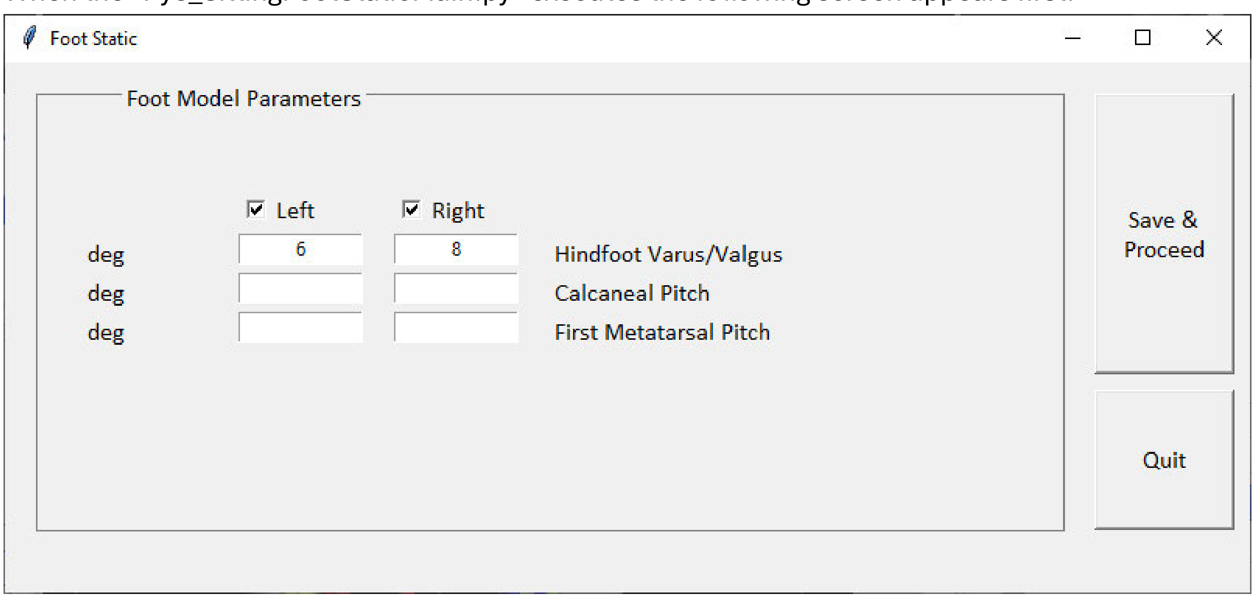
    c. Label only the feet. (Note that this means you don’t have to worry about seeing pelvis or other markers). <u>DON’T need to label markers on Malleoli either.</u> Example labeling shown below:



4) Run the following commands in the 'Python3_SittingFoot_Static_BF' pipeline for this trial
   a. The Sitting Foot Static (Run Python Operation)
      - NOTE: the operation needs to point to the following python file: "Py3_SittingFootStaticMain.py"
   b. Export C3D
   c. Save trial

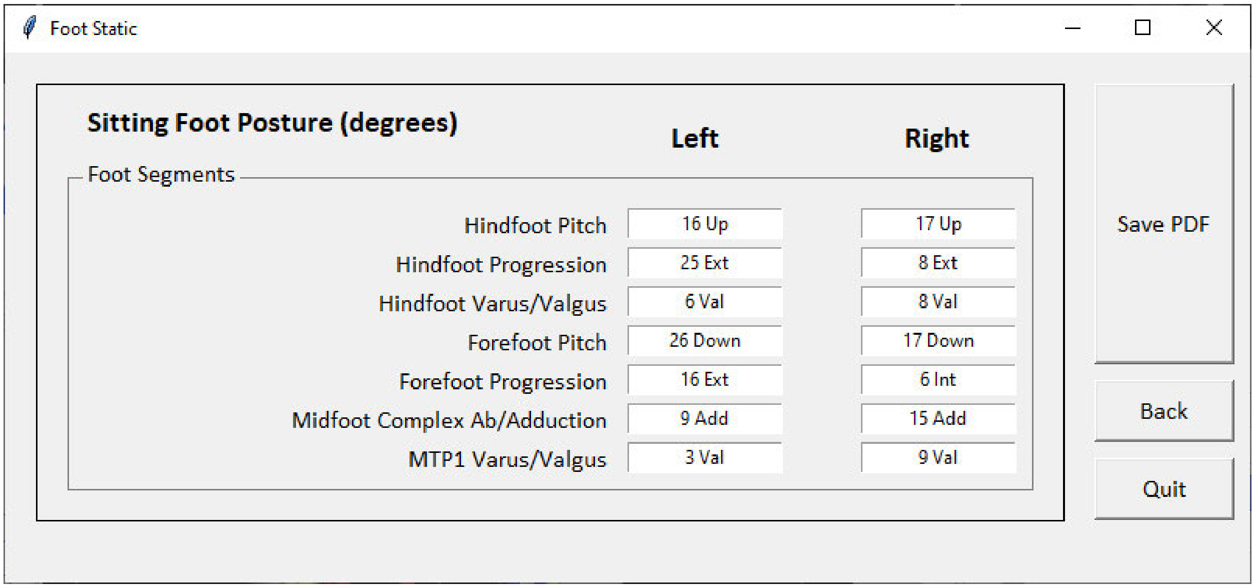


5) When the "Py3_SittingFootStaticMain.py" executes the following screen appears first:



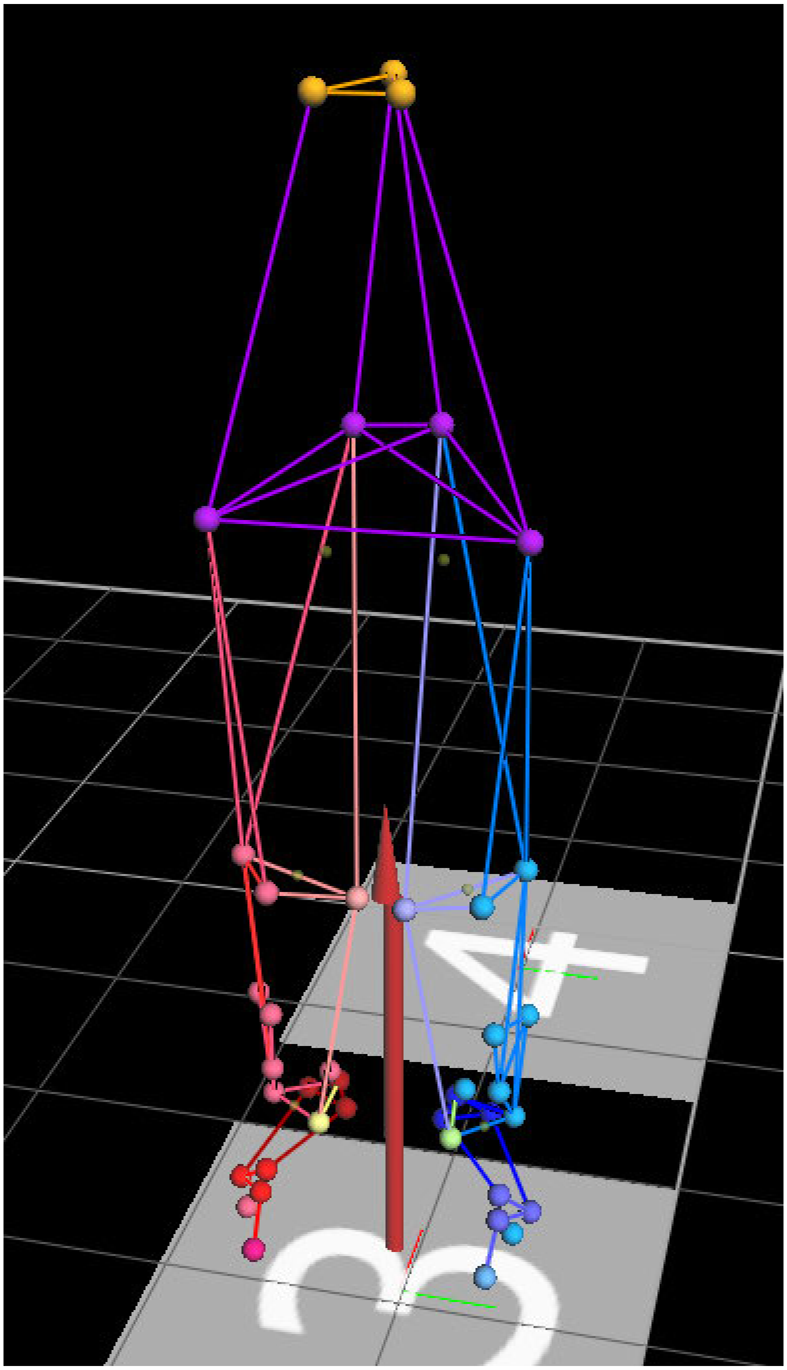
Left and Right should be checked and appropriate measures entered. Calcaneal pitch will need to be updated, and the process repeated in post-processing, if you plan to use radiograph measurements and they have not been calculated before collection.

a. Click save and proceed and the following quality assurance screen appears:

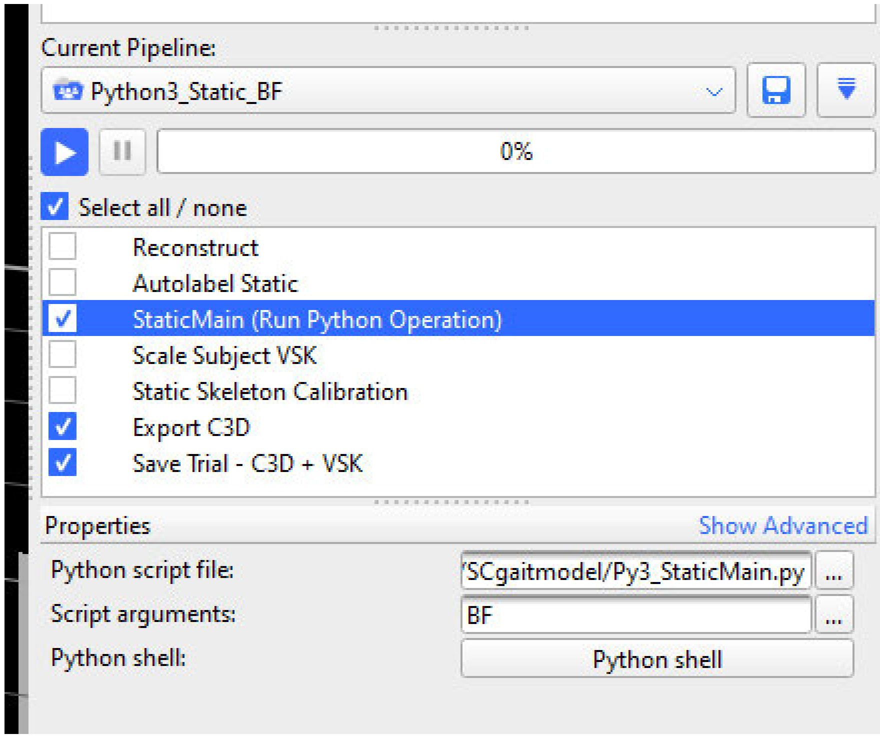


b. Click Save PDF and then Quit

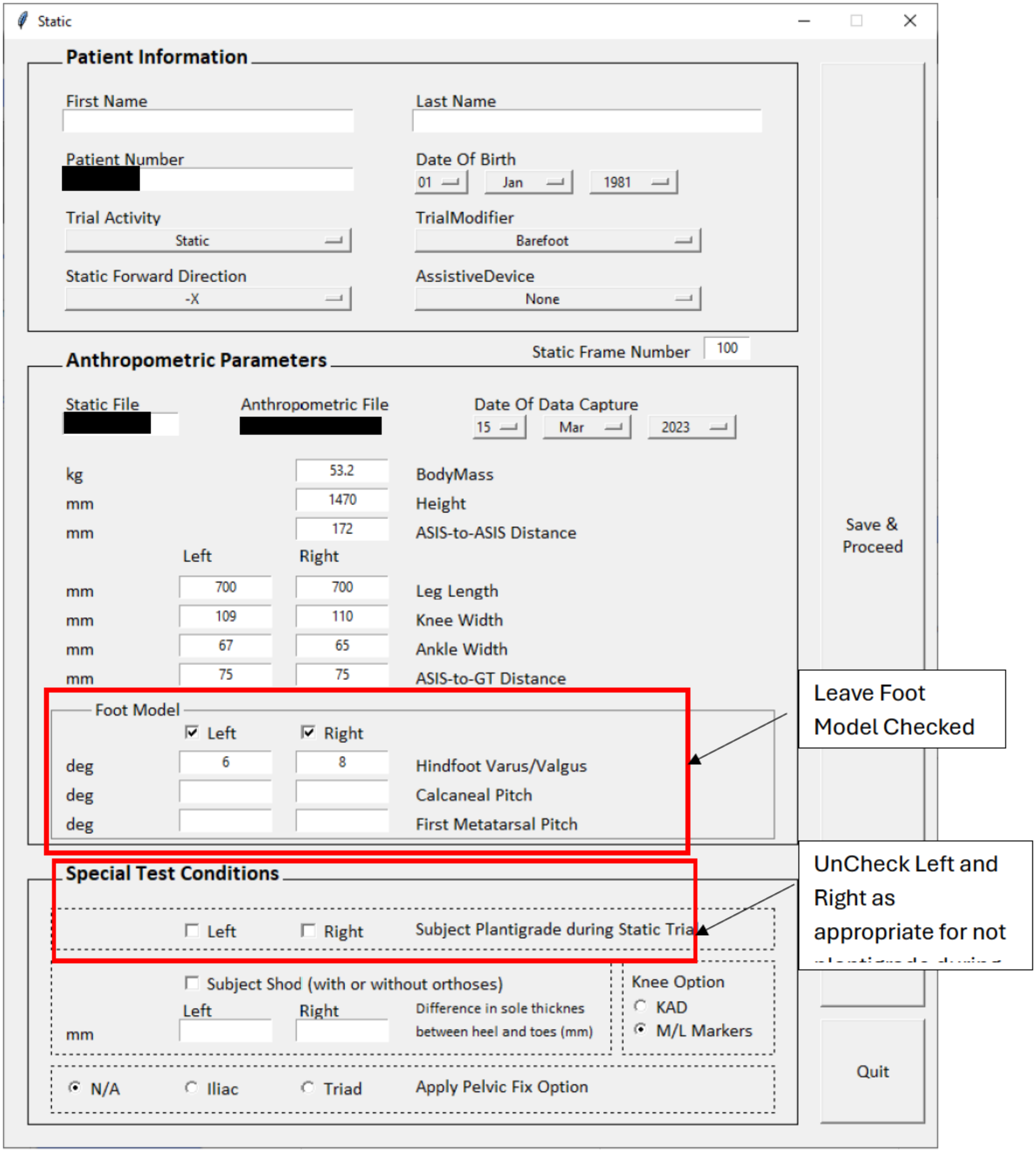
6) Remove foot calibration markers EXCEPT 2nd3rd_MT_Head
7) Place typical Lower Extremity marker set on subject
   a. BE SURE to include 2nd3rd_MT_Head and Heel markers (Leave anatomical 5mm hemisphere from foot model as 2nd3rd_MT_Head; for Heel marker, add physical marker above PCAL that is anatomic heel same height as toe marker).
      - Note: If your PCAL marker is already at this position, you can use it as the heel marker
   b. Make sure both Heel markers are visible
   c. Capture data: Static
   d. Attach Labelling Skeleton Template 'SMACNet Foot Model no KAD FootNotFlat 2ndStepNew
   e. Use the following pipeline: Python3_Static_BF
   f. Reconstruct and label the standing static as if using the standard model
   g. Label everything following example shows labeled stick figure:



h. Run pipeline command that executes Py3_StaticMain.py



i. When the “Py3_StaticMain.py” executes the following screen appears first

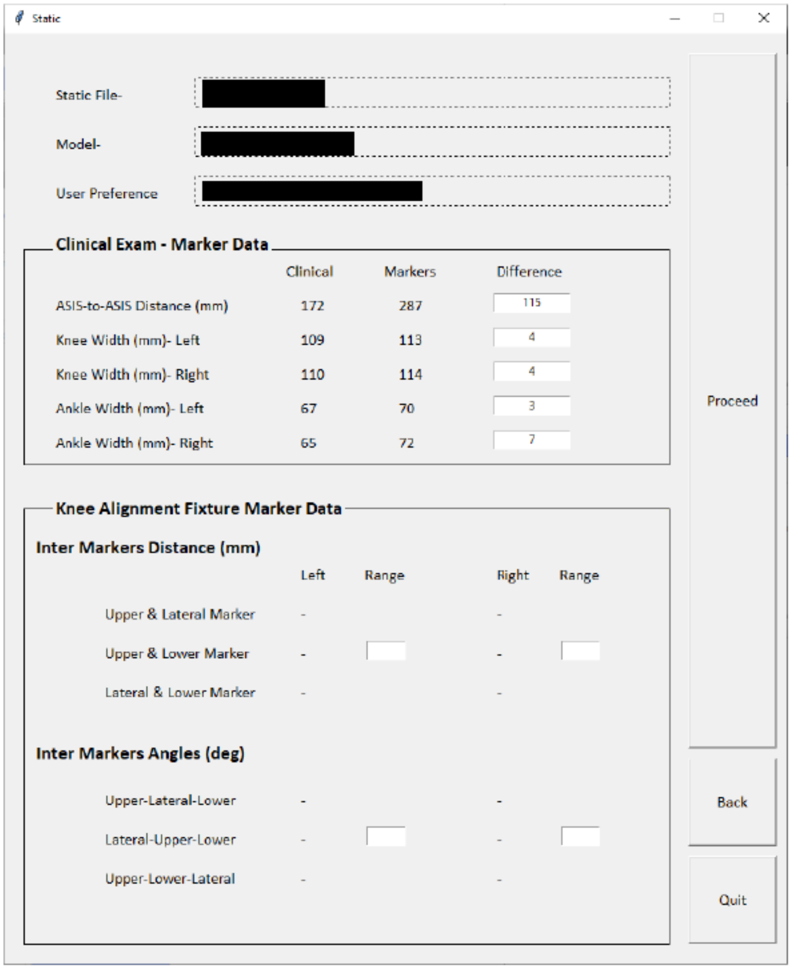


IMPORTANT NOTES:

a. Leave Foot Model checked
b. UNCHECK Subject Plantigrade during Static Trial

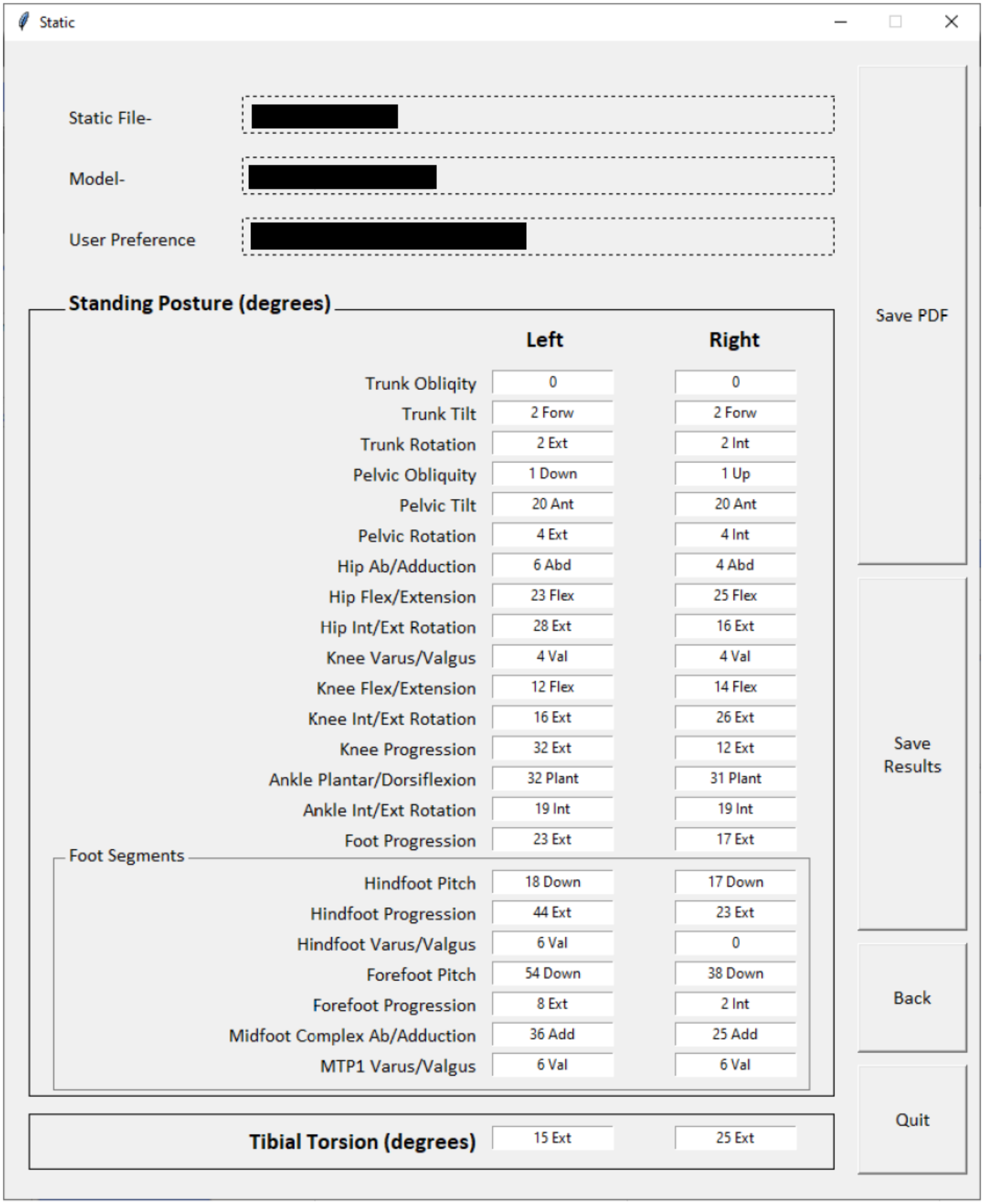
j. Click Save and Proceed button

k. Next Screen appears:

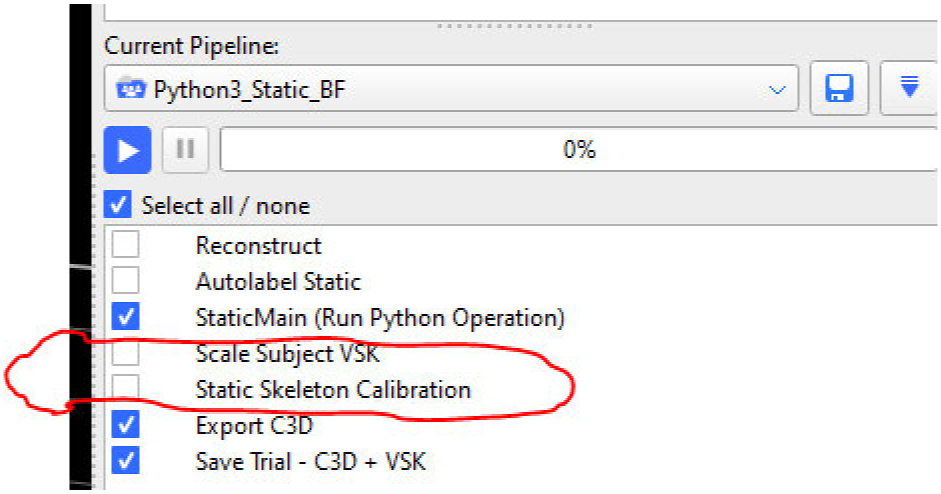


l. Click Proceed button

m. The following quality assurance screen appears:



n. Click Save Results and then Quit
o. In Python3_Static_BF pipeline run Scale Subject VSK and Static Skeleton Calibration to setup autolabeling:



p. Save C3D/save the subject file

8) Remove 2nd/3rd_MT_Head and Heel markers
   a. Leave PCAL marker if used as the heel marker
9) Collect dynamic trials
10) Process dynamic trials SAME AS TYPICAL)
   a. Use ‘Python3_Dynamic_BF‘ Pipeline as usual.